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Substantive Statistics in the Implementation of California's Racial Justice Act:
A Case Study and Recommendations

Introduction

- California's Racial Justice Act (RJA) allows defendants to file a claim if they believe racial bias has influenced their case [1].
- Racial disparities can occur due to prosecutorial discretion during charging (e.g., *wobblers* and *enhancements*) and sentencing (e.g., *plea bargains*) .
- There are four different types of claims, two of which are statistics based: **a(3)** claims address charging disparities and **a(4)** claims address more severe sentencing.
- Both types of claims require comparing only *similarly situated* defendants who engaged in *similar conduct*, though the comparators need not be identical.
- While these claims inherently invoke statistics, there is little guidance on best practices for data selection and what methods are appropriate.
- These claims are also statistically interesting: they often involve the full population of relevant cases and may have small datasets.

Translating the Legal Question to Statistics

Claims process: two stages with different burdens of proof

- *Prima Facie* – "substantial likelihood" or more than a mere possibility
- *Evidentiary hearing* – preponderance of evidence or more likely than not

The legal question: are there racial disparities in charging or sentencing among similarly situated defendants who engaged in similar conduct?

Statistical approaches: over h outcomes and i cases in the dataset

- Linear regression: can be calibrated parametrically or with permutations

$$\mathrm{Y_i^h} = \beta_0^h + \beta_1^h \mathrm{Z_i} + \varepsilon_i$$
$$\mathrm{H_0^h}: \beta_1^h = 0, \qquad \mathrm{H_a^h}: \beta_1^h \neq 0$$

- Permutation tests and non-parametric combination of tests:

$$\mathrm{H_0^h}: \mathrm{Y_i^h}(1) = \mathrm{Y_i^h}(0), \textbf{ for all } h, i, \qquad \mathrm{H_a^h}: \frac{1}{\mathrm{N}}\sum_{\mathrm{i}} \mathrm{Y_i^h}(1) - \mathrm{Y_i^h}(0) > 0, \textbf{ for some } \mathrm{h, i}$$

$$\mathrm{H_0} = \bigcap_{\mathrm{h}\in[\mathrm{H}]} \mathrm{H_0^h} \qquad \mathrm{H_a} = \bigcup_{\mathrm{h}\in[\mathrm{H}]} \mathrm{H_a^h}$$

Table 1: Closed and pending RJA cases with appealed cases starred and one of our cases highlighted [2]

Case	Methods	Court	Prima Facie	Evidentiary Hearing
Mosby v. Superior Court (2024)*	Descriptive Statistics	Court of Appeal	Successful	In progress
People v. Jenkins (2025)*	Logistic Regression	Superior Court	Successful	Unsuccessful
People v. Harris (2021)*	Risk, Inferential Statistics	Superior Court	Unsuccessful	N/A
People v. Jimenez (2025)*	Descriptive Statistics	Superior Court	Unsuccessful	N/A
People v. Windom (2023)	Relative Risk, Odds Ratio	Superior Court	Successful	Successful
People v. Henderson (2025)*	Logistic Regression	Court of Appeal	Unsuccessful	N/A
In re Nelson (2023)*	Logistic Regression	Superior Court	Unsuccessful	Unsuccessful
People v. Grubbs (2026)	**Relative Risk, Permutation Tests, NPC**	**Superior Court**	**Successful**	**Successful**

Case Study

- Consider a real-world case where a Latino defendant is being charged with felony attempted murder and a gun injury enhancement after a physical altercation where they shot the victim.
- Compared to individuals who engaged in similar conduct and are similarly situated :

1. Are Latino defendants charged with felonies more often?
2. Are Latino defendants charged with enhancements more often?
3. Do Latino defendants face worse outcomes (i.e., prison, probation, dismissal)?
4. Do Latino defendants face longer prison sentences?

Table 2: Charging disparities permutation, NPC, and linear regression test statistics and p-values.

	Felony	Enhancement	Omnibus
Permutation Test Statistic	0.36	0.36	16.07
Permutation P-Value	0.02**	0.02**	0.01**
LR Coefficient β	1.73	1.73	-
LR Parametric P-value	0.02**	0.02**	-
LR Permutation P-value	0.04**	0.04**	-

Table 3: Outcome disparities permutation, NPC, and linear regression test statistics and p-values.

	Prison	Probation	Dismissal	Prison Sentence Length	Omnibus
Permutation Test Statistic	0.40	0.21	0.19	72.33	26.58
Permutation P-Value	0.01**	0.13	0.08*	0.003***	0.003***
LR Coefficient β	2.14	-1.35	-21.55	72.33	-
LR Parametric P-value	0.02**	0.14	1.00	0.01**	-
LR Permutation P-value	0.02**	0.19	0.18	0.00***	-

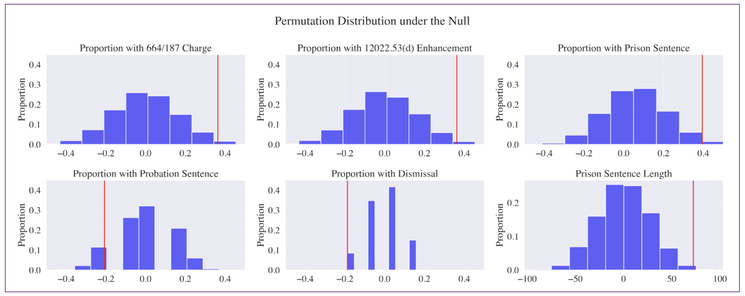


Figure 1. Permutation distributions induced by the null hypotheses, with observed test statistics marked by a red vertical line.

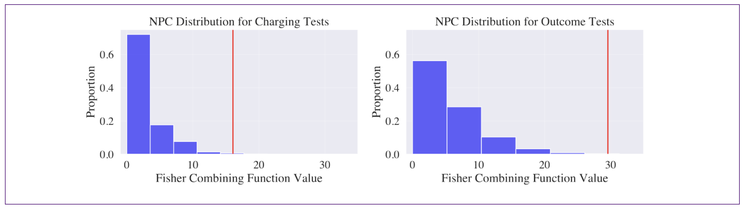


Figure 2. NPC distributions induced by the null hypothesis with observed test statistics marked by a red vertical line

Recommended Methods

Dataset selection

- Data should be selected alongside legal experts to address *similarly situated* and *similar conduct.*
- Data should be filtered prior to analysis to avoid Simpson's paradox

Permutation Tests [3]:

- Define the test statistic as the difference of the averages between two races:

$$\mathrm{T(Y, X)} = \left(\frac{1}{\mathrm{A}} \sum_{j:\, \mathrm{X_j}=\mathrm{Latino}} \mathrm{Y_j}\right) - \left(\frac{1}{\mathrm{B}} \sum_{j:\, \mathrm{X_j}=\mathrm{white}} \mathrm{Y_j}\right)$$

- Let $\mathrm{T}(\mathrm{Y}, \widetilde{\mathrm{X}}_\mathrm{r})$ be the test statistic on replicate $\mathrm{r} \in [\mathrm{R}]$, with $\widetilde{\mathrm{X}}_\mathrm{r} \sim \mathrm{Unif}\{\mathrm{g(x)}\colon\ \mathrm{g} \in \mathcal{G}\}$ where $\mathcal{G}$ is the permutation group
- The P-value is then:

$$p = \frac{1 + \#\{\mathrm{r}\colon\ \mathrm{T}(\mathrm{Y}, \widetilde{\mathrm{X}}_\mathrm{r}) \geq \mathrm{T}(\mathrm{Y}, \mathrm{X})\}}{1 + R}$$

Non-Parametric Combination of Tests (NPC) [4]:

- Test over $\mathrm{h} = 1 \ldots \mathrm{H}$ hypothesis with corresponding test statistics $\mathrm{T_h}$ and outcomes of interest $\mathrm{Y^h}$
- For each of $\mathrm{r} = 1 \ldots \mathrm{R}$ randomizations, we get a test statistic for test, $\mathrm{T_h}(\mathrm{Y^h}, \widetilde{\mathrm{X}}_\mathrm{r})$, as well as a corresponding p-value, $p_{\mathrm{h,r}}$

$$p_{\mathrm{h}} = \frac{1 + \#\{\mathrm{r}\colon\ \mathrm{T_h}(\mathrm{Y^h}, \widetilde{\mathrm{X}}_\mathrm{r}) \geq \mathrm{T_h}(\mathrm{Y^h}, \mathrm{X})\}}{1 + R}$$
$$p_{\mathrm{h,r}} = \frac{1 + \#\{\mathrm{s}\colon\ \mathrm{T_h}(\mathrm{Y^h}, \widetilde{\mathrm{X}}_\mathrm{s}) \geq \mathrm{T_h}(\mathrm{Y^h}, \widetilde{\mathrm{X}}_\mathrm{r})\}}{1 + R}$$

- For each randomization, we can combine p-values using a function, e.g., Fisher's:

$$f_0(p_1, \ldots, p_\mathrm{H}) = -2 \sum_{\mathrm{h}\in[\mathrm{H}]} ln\, p_\mathrm{h}\,; \quad f_\mathrm{r}(p_{1,\mathrm{r}}, \ldots, p_{\mathrm{H,r}}) = -2 \sum_{\mathrm{h}\in[\mathrm{H}]} ln\, p_{\mathrm{h,r}}$$

- The omnibus p-value is then:

$$p_{NPC} = \frac{1 + \#\{\mathrm{r}\colon\ f_\mathrm{r} \geq f_0\}}{1 + R}$$

Discussion

- We have successfully provided statistical expert consulting on two cases resulting in favorable outcomes (People v. Grubbs 2026 and the anonymous case study).
- Descriptive statistics and permutation-based test have shown success in real RJA claims while logistic regression-based analyses have yet to succeed.
- While linear regression is appealing, it should be used with caution and calibrated with the permutation distribution.

References

[1] California State Bill (2022), 'Assembly Bill No. 256'
[2] Romo, L. M. (2025), 'The Disparity in Litigating Racial Disparity Claims: The Need for California Courts to Articulate a Framework for Assessing Racial Justice Act Challenges to Charging, Conviction, and Sentencing', 65(1), 229
[3] A. K. Glazer & P. B. Stark. Fast Conservative Monte Carlo Confidence Sets. Journal of Computational and Graphical Statistics. 2025
[4] Pesarin, Fortunato and Salmaso, Luigi. "Permutation tests for complex data: Theory, applications and software". Chapter 5. John Wiley & Sons, 2010.